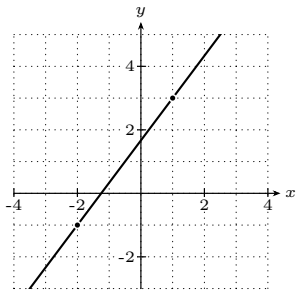
Algebra Review: Unit 3 — Linear Equations & Slope

Station 1 of 4: Finding and Interpreting Slope

Name: _____ Date: _____

Write slopes as simplified fractions or integers. Write "undefined when appropriate. Diagrams are not drawn to scale.

1. Find the slope of the line shown. Use the marked points.

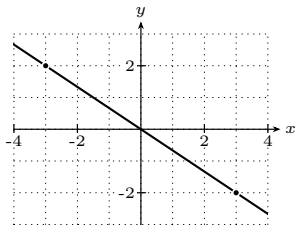


2. Find the slope of the line through $(-2, 5)$ and $(4, -1)$.

3. Find the slope of the line through $(0, -3)$ and $(6, -3)$.

4. Find the slope of the line through $(4, 1)$ and $(4, 7)$.

5. Find the slope of the line shown. Use the marked points.



6. The line through $(2, k)$ and $(5, 10)$ has slope 3. Find k.

7. A line passes through $(0, 5)$, $(2, 11)$, and $(4, 17)$. What is its slope?

8. Find the slope of the line $4x - 2y = 8$.

9. A plane descends from 30,000 ft to 18,000 ft in 15 minutes. Find the rate of change in feet per minute.

10. Are the points $(-1, -3)$, $(1, 1)$, and $(4, 7)$ on the same line? Explain using slope.

Algebra Review: Unit 3 — Linear Equations & Slope

Station 2 of 4: Slope-Intercept Form

Name: _____ Date: _____

Write equations in the form $y = mx + b$ unless told otherwise. Diagrams are not drawn to scale.

1. Identify the slope and y-intercept: $y = -3x + 7$

2. Identify the slope and y-intercept: $y = \frac{2}{5}x - 4$

3. Rewrite $6x + 2y = 10$ in slope-intercept form and identify m and b.

4. Write the equation of the line with slope 4 and y-intercept -2.

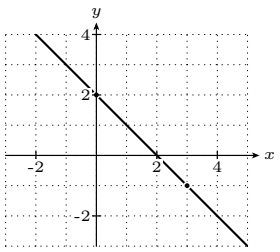
5. Write the equation of the line with slope $-\frac{1}{2}$ and y-intercept 6.

6. Write the equation of the line with slope 3 that passes through $(2, 5)$.

7. Write the equation of the line through $(-1, 4)$ and $(3, -4)$.

8. Write the equation of the line through $(0, -1)$ and $(4, 5)$.

9. Write the equation of the line shown in slope-intercept form.



10. A taxi charges a \$3 base fare plus \$2 per mile. Write an equation for the cost y of a ride of x miles, then find the cost of a 12-mile ride.

Algebra Review: Unit 3 — Linear Equations & Slope

Station 3 of 4: Point-Slope and Standard Form

Name: _____ Date: _____

Use the form requested. In standard form $Ax + By = C$, use integer coefficients with $A > 0$.

1. Write in point-slope form: slope 2 through $(3, -4)$.

2. Write in point-slope form: slope $-\frac{1}{3}$ through $(-6, 1)$.

3. Write in point-slope form the line through $(1, 2)$ and $(4, 8)$ (use the point $(1, 2)$).

4. Rewrite $y - 5 = 3(x - 2)$ in slope-intercept form.

5. Rewrite $y + 1 = -2(x - 4)$ in slope-intercept form.

6. Write $y = 2x - 5$ in standard form.

7. Write $y = -\frac{3}{4}x + 2$ in standard form.

8. Find the x- and y-intercepts of $3x + 5y = 15$.

9. Find the x- and y-intercepts of $2x - 3y = 12$.

10. Write, in standard form, the equation of the line through $(2, 1)$ with slope -3.

Algebra Review: Unit 3 — Linear Equations & Slope

Station 4 of 4: Parallel and Perpendicular Lines

Name: _____ Date: _____

Remember: parallel lines have equal slopes; perpendicular lines have slopes that are opposite reciprocals.

1. Find the slope of a line parallel to $y = 4x - 1$.

2. Find the slope of a line perpendicular to $y = 4x - 1$.

3. Find the slope of a line perpendicular to a line with slope $-\frac{2}{3}$.

4. Write the equation of the line through $(0, 3)$ that is parallel to $y = -2x + 5$.

5. Write the equation of the line through $(4, 1)$ that is perpendicular to $y = 2x - 3$.

6. Write the equation of the line through $(1, -2)$ that is parallel to $3x - y = 4$.

7. Are $y = 2x + 1$ and $4x - 2y = 6$ parallel, perpendicular, or neither?

8. Are $y = 3x + 2$ and $x + 3y = 9$ parallel, perpendicular, or neither?

9. Are $2x + y = 5$ and $x + 2y = 4$ parallel, perpendicular, or neither?

10. Write the equation of (a) the horizontal line through $(-3, 7)$ and (b) the vertical line through $(5, -2)$.

Algebra Review: Unit 3 — Linear Equations & Slope

Answer Key

Station 1: Finding and Interpreting Slope

1. $m = \frac{4}{3}$ *points* $(-2, -1)$ *and* $(1, 3)$
2. $m = -1$
3. $m = 0$
4. undefined
5. $m = -\frac{2}{3}$ *points* $(-3, 2)$ *and* $(3, -2)$
6. $k = 1$ $\frac{10 - k}{3} = 3$
7. $m = 3$
8. $m = 2$ $y = 2x - 4$
9. $-\frac{12{,}000}{15} = -800$ ft/min
10. Yes. slope $= \frac{4}{2} = \frac{6}{3} = 2$ for both pairs

Station 2: Slope-Intercept Form

1. $m = -3,\ b = 7$
2. $m = \frac{2}{5},\ b = -4$
3. $y = -3x + 5;\ m = -3,\ b = 5$
4. $y = 4x - 2$
5. $y = -\frac{1}{2}x + 6$
6. $y = 3x - 1$ $5 = 3(2) + b$
7. $y = -2x + 2$ $m = \frac{-8}{4} = -2$
8. $y = \frac{3}{2}x - 1$
9. $y = -x + 2$ $b = 2$*;* $m = \frac{-1 - 2}{3 - 0} = -1$
10. $y = 2x + 3$; \$27

Station 3: Point-Slope and Standard Form

1. $y + 4 = 2(x - 3)$
2. $y - 1 = -\frac{1}{3}(x + 6)$
3. $y - 2 = 2(x - 1)$
4. $y = 3x - 1$
5. $y = -2x + 7$
6. $2x - y = 5$
7. $3x + 4y = 8$
8. x-int: $(5, 0)$; y-int: $(0, 3)$
9. x-int: $(6, 0)$; y-int: $(0, -4)$
10. $3x + y = 7$

Station 4: Parallel and Perpendicular Lines

1. 4
2. $-\frac{1}{4}$
3. $\frac{3}{2}$
4. $y = -2x + 3$
5. $y = -\frac{1}{2}x + 3$ $1 = -\frac{1}{2}(4) + b$
6. $y = 3x - 5$
7. Parallel *second line:* $y = 2x - 3$
8. Perpendicular *slopes* 3 *and* $-\frac{1}{3}$
9. Neither *slopes* -2 *and* $-\frac{1}{2}$
10. (a) $y = 7$ (b) $x = 5$